

Feature Importance Report

Introduction:

This section analyzes the relative importance of each feature in predicting student depression risk. Feature importance was extracted from the best-performing model (Random Forest Classifier). The higher the importance percentage, the more influential the feature in model predictions.

Feature Importance Ranking:

Feature	Importance (%)
Stress Level	35.00
Sleep Quality	25.00
Study Hours	15.00
Anxiety Level	15.00
Social Activity	10.00

Interpretation:

Stress Level emerged as the most critical predictor of depression risk, contributing 35% to the model's decisions. Sleep Quality and Study Hours also played significant roles, highlighting the impact of lifestyle factors on student mental health. Intervention strategies should prioritize stress reduction and sleep quality improvement.